



McMASTER UNIVERSITY

Ebola Transmission Modelling Based on SIR Model

AUTHOR INFORMATION

Submitted by:

Jiajun Zhang (400359213)
Zhixuan Li (400612826)
Jingchen Zou (400359056)
Chengdai Xu (400397089)

Instructor:

Prof. Sicheng Zhao

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1 Introduction

1.1 Background Information

Known for its high pathogenicity and lethality, the Ebola virus was first detected in 1976, when two outbreaks occurred in the Nzara region of what is today South Sudan and in the small town of Yambuku in the Democratic Republic of Congo. The latter was closer to the Ebola River, hence the name “Ebola” (Feldmann & Geisbert, 2011). Since the first discovery of Ebola in 1976, the vast majority of cases and outbreaks have been concentrated in Africa (Amundsen, 1998), with a total of 36 outbreaks occurring in six countries (Aurelie et al., 2017). The most recent large-scale outbreak lasted for two years in the Democratic Republic of Congo, extending from 2018 to 2020((WHO), 2021).

According to the World Health Organization (WHO) 2021, the Leschenault’s fruit bat is considered as a possible carrier host for the Ebola virus. Studies have hypothesized that the virus is transmitted to humans via animals in forest ecosystems (Olivall et al., 2013). Ebola virus is spread among people mainly through contact with blood or body fluids of infected people or people who have died from the infection. In addition, contact with objects contaminated with these bodily fluids (e.g., blood, feces, or vomit) can also lead to transmission (Peters & LeDuc, 1999). According to WHO ((WHO), 2021), the diverse routes of transmission of Ebola virus, including sexual contact and mother-to-child transmission, make it a major challenge at the public health level and increase the difficulty of virus control and management.

Preventing and controlling the spread of Ebola virus is critical but challenging. Modeling Ebola virus transmission plays a central role in disease prevention. For example, the SIFR model and its derivatives have been applied to simulate the transmission dynamics of Ebola to help predict the trend of the epidemic and evaluate the effective-

ness of prevention and control strategies, thus providing valuable references for policy formulation. Our study analyzed such models in depth to examine the transmission mechanism of Ebola virus in a socio-cultural context. In particular, we examined how personal interactions and community activities (e.g., rituals) influence the spread of the epidemic. Traditional funeral practices in many countries often involve direct contact with the deceased, which has a significant impact on disease transmission. Inappropriate burial practices have played a significant role in the spread of Ebola in Africa, especially in the 2014 Ebola outbreak in Guinea, where these practices are thought to have accounted for 68% of the cases (Chan, 2014). To further refine Ebola management strategies, we plan to explore the following topics in depth during the project.

1.2 Research Questions

1. What parameter values can we use to monitor the infection?
2. How do funerals contribute to the spread of the Ebola virus within populations?
3. What is the impact of public health interventions in reducing transmission in funerals on epidemic control?

2 Base Model

2.1 Model Outline

The purpose of this model is to monitor the infection and spread of EVD. We will employ an SIFR model, which tracks the rate of change across four distinct groups or populations: individuals who are susceptible, infectious/infected, attending funerals, and deceased/removed. Through these simulations, we aim to observe the variations in

each of these curves and evaluate how the epidemic's progression (i.e., the shape of the curves) could be influenced by various interventions. To start with the base model, we will examine the different populations and define the relevant parameters.

2.2 Base Model Assumptions

The conditions of the real world is very complicated, thus we should make some assumptions to simplify the base model:

- Assume there is no population fluctuation, considering a fixed population;
- Assume all people are equally treated;
- Assume all infected individuals will die;
- Assume all individuals who have completed as the healthy individuals;
- Assume all the rates are fixed, not affected by other factors;
- Assume all the newly-born are healthy;
- Assume each individual moves linearly through the SIFR model.
- Assume the natural death will not enter the funeral population.

2.3 State Variables and Parameters

State Variables:

- S : Fraction of the population that is susceptible (Percentage)
- I : Fraction of the individual that is infectious and not hospitalized (Percentage)
- F : Fraction of the individual that is dead individuals whose funeral ceremonies are not yet completed (Percentage)

- R : Fraction of the individual that is no longer infectious (Healthy) (Percentage)

Parameters:

- γ : Rate of death after infection within a unit of time (Percentage)
- β : Effective connect rate between an infectious and susceptible individual within a unit of time (Transmission per capita rate)
- β_F : Transmission rate between susceptible and dead but unburied individuals within a unit of time
- μ : Both rate of birth and rate of death within a unit of time
- σ : Burial rate for deaths to funerals within a unit of time

2.4 Modeling Choosing

To analyze the fluctuations in the number of Ebola virus infections, the modes of transmission, and effective prevention strategies, we employ the SIFR model, a widely used framework in modern epidemiology. This model focuses on three primary groups: susceptible individuals (S), infectious individuals (I), and deceased individuals associated with funeral-related transmission (F). The analysis particularly emphasizes the role of funerals in the spread of the disease. Below are the process flowchart and the corresponding mathematical formulas for this model.

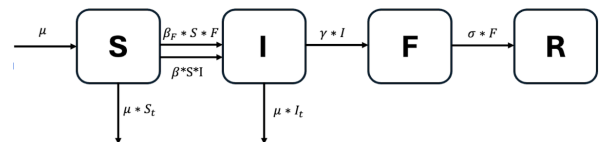


Figure 1: SIFR Model Process Flowchart

Then we can combine all the information we have and get our mathematical

model, it is a system which contains four equations

$$\begin{aligned} S_{t+1} &= S_t - \beta S_t I_t - \beta_F S_t F_t + \mu - \mu S_t \\ I_{t+1} &= I_t + \beta I_t S_t + \beta_F S_t F_t - \gamma I_t - \mu I_t \\ F_{t+1} &= F_t + \gamma I_t - \sigma F_t \\ R_{t+1} &= R_t + \sigma F_t \end{aligned}$$

For susceptible individuals (S_{t+1}), the population decreases due to contact with infectious individuals ($-\beta S_t I_t$) and funeral-related transmission ($-\beta_F S_t F_t$). Births contribute to an increase in the susceptible population (μ), while natural deaths reduce it ($-\mu S_t$). For infectious individuals (I_{t+1}), the population grows through new infections caused by interactions between susceptible and infectious individuals ($\beta S_t I_t$) and funeral-related transmission ($\beta_F S_t F_t$), but decreases as individuals die from the infection ($-\gamma I_t$) or succumb to natural causes ($-\mu I_t$). For the funeral-related deceased (F_{t+1}), their number increases as infectious individuals die (γI_t), but decreases as funeral transmission ceases due to proper handling of the deceased ($-\sigma F_t$). Finally, for recovered individuals (R_{t+1}), their number increases as funeral-related deceased are properly handled and no longer infectious (σF_t).

2.5 Mathematical Analysis for Equilibria and its Stability

We use the theory of equilibria, which has the equation:

$$\begin{aligned} S_{t+1} &= S_t = S^* \\ F_{t+1} &= F_t = F^* \\ I_{t+1} &= I_t = I^* \\ R_{t+1} &= R_t = R^* \end{aligned}$$

And thus we can compute and find the equilibrium point of this system of model. It should be

$$S^* = 1, \quad I^* = 0, \quad F^* = 0, \quad R^* = R^*$$

In conclusion, the model suggests that, over time, the system will asymptotically

approach a disease-free equilibrium. This equilibrium represents a state where no new Ebola infections occur, no deaths are attributed to the virus, and the population transitions entirely to the susceptible compartment. However, due to the inherent limitations of the model, it does not provide an estimation of the cumulative number of deaths caused by Ebola or the associated burials.

While the theoretical analysis indicates convergence to this equilibrium, the actual dynamics of the system may deviate due to non-linear interactions or parameter variations. To rigorously analyze the stability and behavior of the system near equilibrium, it is essential to compute and evaluate the corresponding Jacobian matrix. Let us proceed to derive this matrix and examine its implications. We construct these functions from the model:

$$\begin{aligned} f_1 &= S - \beta SI - \beta_F SF + \mu - \mu S \\ f_2 &= I + \beta SI + \beta_F SF - \gamma I - \mu I \\ f_3 &= F + \gamma I - \sigma F \\ f_4 &= R + \sigma F \end{aligned}$$

The Jacobian Matrix is given by:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial S} & \frac{\partial f_1}{\partial I} & \frac{\partial f_1}{\partial F} & \frac{\partial f_1}{\partial R} \\ \frac{\partial f_2}{\partial S} & \frac{\partial f_2}{\partial I} & \frac{\partial f_2}{\partial F} & \frac{\partial f_2}{\partial R} \\ \frac{\partial f_3}{\partial S} & \frac{\partial f_3}{\partial I} & \frac{\partial f_3}{\partial F} & \frac{\partial f_3}{\partial R} \\ \frac{\partial f_4}{\partial S} & \frac{\partial f_4}{\partial I} & \frac{\partial f_4}{\partial F} & \frac{\partial f_4}{\partial R} \end{bmatrix}$$

Substituting the partial derivatives:

$$J_{(1,0,0,R^*)} = \begin{bmatrix} 1 - \mu & -\beta & -\beta_F & 0 \\ 0 & 1 + \beta - \mu - \gamma & \beta_F & 0 \\ 0 & \gamma & 1 - \sigma & 0 \\ 0 & 0 & \sigma & 1 \end{bmatrix}$$

We then solve for the eigenvalues of this matrix to evaluate system stability. Let λ represent the eigenvalue:

$$\det(J - \lambda I) = 0$$

We expand this determinate and get that:

$$\det(J - \lambda I) = \begin{vmatrix} 1 - \mu - \lambda & -\beta & -\beta_F & 0 \\ 0 & 1 + \beta - \mu - \gamma - \lambda & \beta_F & 0 \\ 0 & \gamma & 1 - \sigma - \lambda & 0 \\ 0 & 0 & \sigma & 1 - \lambda \end{vmatrix}$$

Solving this equation gets eigenvalues:

$$\begin{aligned}\lambda_1 &= 1, & \lambda_2 &= 1 - \mu \\ \lambda_3 &= \frac{-b + \sqrt{b^2 - 4ac}}{2a} \\ \lambda_4 &= \frac{-b - \sqrt{b^2 - 4ac}}{2a}\end{aligned}$$

Where a,b,c should be represented by:

$$\begin{aligned}a &= 1 \\ b &= \sigma - 2 - \beta + \mu + \gamma \\ c &= (1 + \beta - \mu - \gamma)(1 - \sigma) - \gamma\beta_F\end{aligned}$$

In dynamic systems, the magnitude of eigenvalues reflects the stability of the system. Specifically, the dominant eigenvalue (the eigenvalue with the largest absolute value) determines the long-term behavior of the system

$$\lambda \begin{cases} < 1 & \text{Stable (disease-free)} \\ = 1 & \text{Critical (trajectory uncertain)} \\ > 1 & \text{Unstable (disease persists)} \end{cases}$$

Therefore, the dominant eigenvalue is a critical metric for evaluating the stability of an epidemic model and the effectiveness of control measures.

2.6 Some possible scenarios

2.6.1 Scenario Where the Dominant Eigenvalue Equals 1

Model Behavior: When the dominant eigenvalue is exactly 1, the trajectory of the model (system behavior) cannot be easily determined through mathematical analysis. The system may exhibit various dynamic patterns, such as sustained oscillations or stabilization at a certain state.

Implication: In this scenario, it is uncertain whether the disease will be completely eradicated or continue to persist. Therefore, numerical simulations are required to further study and evaluate how the model evolves over time.

2.6.2 Scenario Where the Dominant Eigenvalue is Greater than 1

Model Behavior: When the dominant eigenvalue is greater than 1, the disease-free equilibrium of the system loses stability, and the epidemic will continue to spread or grow. And this could be the major behaviors for the equilibrium point

Implication: In this case, the model predicts that even with intervention measures, it may be difficult to completely control the spread of the disease.

2.7 Real World Analysis and Policy Suggestion

Based on the characteristics of the Ebola epidemic and model analysis, the following detailed parameter adjustment recommendations aim to effectively control the spread of the disease:

2.7.1 Improve Burial Efficiency (σ)

- **Using more scientific methods:** Ask the professionals for some technical guidance and employ more people who own the professional knowledge.
- **Make Strict Burial Management:** Standardize the actual process of the bury and make some strict policies like don't allow people to touch remains and then accelerate the whole procure for bury.
- **Traditional culture reduction:** Inform the public about the importance of keeping distance and touching. Simply some traditional rituals.

2.7.2 Increase Birth Rates (μ)

- **Bonus Giving for parents:** A big problem for new couples to born a baby is that they aren't afford the expenses. Government should make some policy to attract people and let they know of pros of have a baby.

- **Supporting Maternal and Child Health:** Provide adequate prenatal and postnatal care to ensure the health of newborns and mitigate the negative effects of the epidemic on essential health services.
- **Increase the proficiency:** It is necessary to increase the proficiency of the local hospital. Some people perhaps would be kill by the local dirty environment and not professional doctors. This is going to cause a big deal.

2.7.3 Reduce Transmission (β)

- **Canceling some big activities:** Although it is incredibly significant in Africa, but the association should still focus on the health safety and reduce the chance of touchment among people as much as they can
- **Implement home isolation policy:** Despite some people would regard that as disrespectful to human rights. But we still suggest that government should highlight the life first. And take responsibility to their nationals.
- **Maximize the sense of self-protection:** If some of friends was infected, the most proper approach is that stop all possible way to connect them. However some people won't accept that only be restricted by official departments.

All in all, after doing the analysis of the equilibrium point and their stability, we can decrease the epidemic from making some policy to adjust the factors we used in this model efficiently. The policy we have mentioned aims to keep the dominant eigenvalue below 1. In the context of the Ebola epidemic, these strategies must be tailored to the local cultural context, resource availability, and the government's organizational capacity. Successfully imple-

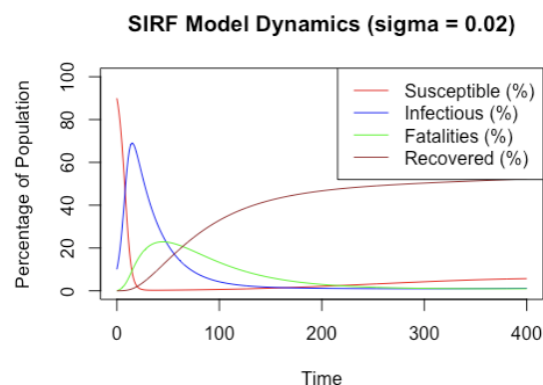
menting these interventions can ensure the long-term goal of zero infections and zero deaths is achieved.

2.8 R Simulation Analysis

To support the summary mentioned above, we can use R simulations to test the change of different variables in our SIRD Model, including susceptible, infectious, fatalities, and recovered population in respect of time-line by adjusting a certain parameter. For all of the following parameter analysis, we focus on performance of various populations during the first 180 days since we can always see the similar trend in the long run (after 180 days).

2.8.1 First R Simulation Analysis

We firstly investigate the relationship between σ and the performance of various populations in the short run. σ represents burial efficiency, it is negatively related to the time from the start of the funeral ceremony to the completion of the burial.



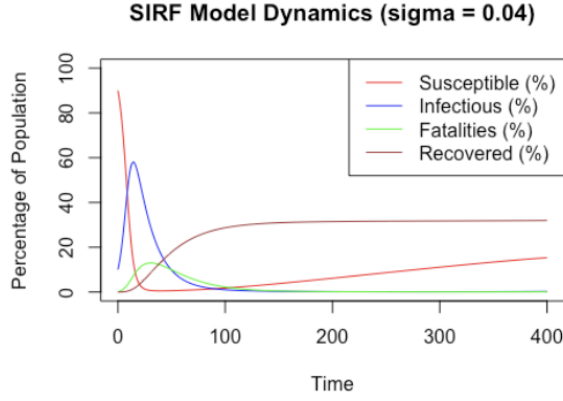


Figure 2: Base Model Simulation Result based on Changes in Burial Efficiency

The result of R Simulation shows that there are less people dying due to attending funerals if burial efficiency rises up. The charts also show that the percentage of the population who got infectious but not hospitalized also decreased. An interesting fact is that the proportion for different populations hardly changes after 150 days after the pandemic exploded. Ideally, if we replace traditional funerals with safer rituals, we can avoid close contact between susceptible populations and virus-carrying bodies and hence effectively restrain Ebola transmission.

2.8.2 Second R Simulation Analysis

Now we use R Simulation to investigate the relationship between beta and percentage change of various populations. Beta represents the infection transmission rate, measuring how fast the pandemic spread.

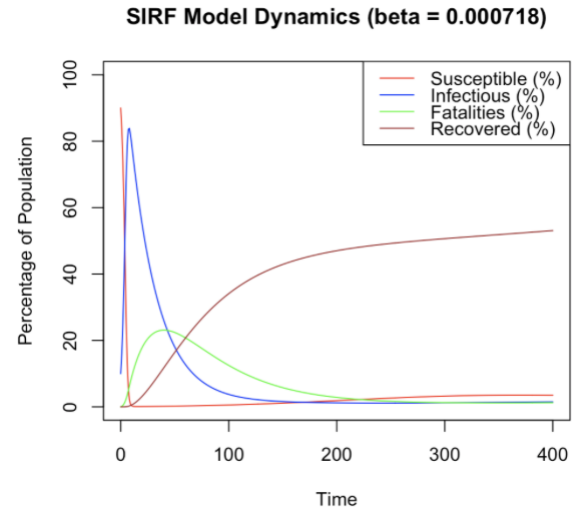
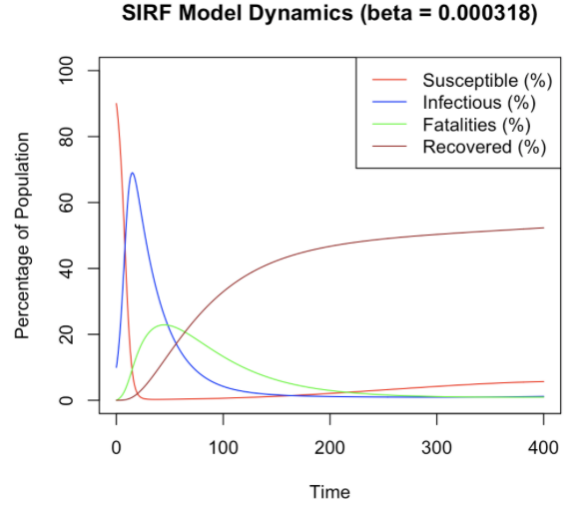


Figure 3: Base Model Simulation Result based on Changes in Infection Transmission Rate

We set $\beta = 0.000318$ for the left graph and $\beta = 0.000718$ for the right one, while keeping other parameters constant. The result of R Simulation shows that the infectious but not hospitalized population is really sensitive to beta change. In the very short run (from day 0 to day 50), the fluctuation of the infectious population gets much stronger if we rise up beta by 0.0004. The peak of infectious population are about 70% and 85% respectively. In addition, there is no big gap in fatalities and recovered proportion under two cases. Unlike the result of sigma R simulation analysis, after day 100, the proportion of various

populations tends to be the same. Based on the observation, we can see that it is essential to control the transmission rate at the very beginning; in the early stages of pandemic, the government should take action immediately, such as introducing strict isolation policy and ensuring proper disinfection in densely populated areas.

3 Extended Model

3.1 Introduction and basical Description on Extended Model

Based on the previous assumptions, state variables and parameters from the part of the base model, we still find some problems with this model. For instance, some of the infected individuals may have died before being buried and are no longer infectious. And our extended model will take this into consideration. Here we introduce a new factor epsilon (ϵ) which indicates the rate of burials without funerals. It reflects strategies such as immediate and safe burial practices or effective isolation measures to prevent further transmission of the virus during funerals from the deceased who are having a funeral. This parameter adjusts the flow from the infected to the deceased who are no longer considered infectious and reduces the flow through the stage of the deceased where funerals are being held. This change allows us to assess the victim after the death of Ebola management interventions, as well as its effect on epidemic overall dynamics.

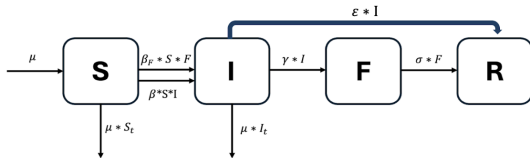


Figure 4: Extension Model Visualization

3.2 Modeling choosing

Then we can combine all the information we have and get our mathematical model, it is a system which contains four equations

$$\begin{aligned} S_{t+1} &= S_t - \beta S_t I_t - \beta_F S_t F_t + \mu - \mu S_t \\ I_{t+1} &= I_t + \beta I_t S_t + \beta_F S_t F_t - \gamma I_t - \mu I_t - \epsilon I_t \\ F_{t+1} &= F_t + \gamma I_t - \sigma F_t \\ R_{t+1} &= R_t + \sigma F_t + \epsilon I_t \end{aligned}$$

For susceptible individuals (S_{t+1}), the population decreases due to contact with infectious individuals ($-\beta S_t I_t$) and funeral-related transmission ($-\beta_F S_t F_t$). Births contribute to an increase in the susceptible population (μ), while natural deaths reduce it ($-\mu S_t$). For infectious individuals (I_{t+1}), the population grows through new infections caused by interactions between susceptible and infectious individuals ($\beta S_t I_t$) and funeral-related transmission ($\beta_F S_t F_t$), but decreases as individuals die from the infection ($-\gamma I_t$) or succumb to natural causes ($-\mu I_t$) and the amount of people who have died immediately ($-\epsilon I_t$). For the funeral-related deceased (F_{t+1}), their number increases as infectious individuals die (γI_t), but decreases as funeral transmission ceases due to proper handling of the deceased ($-\sigma F_t$). Finally, for recovered individuals (R_{t+1}), their number increases as funeral-related deceased are properly handled and no longer infectious (σF_t) and should be added (ϵI_t).

3.3 Mathematical analysis of the extended model

Since we have the same purpose to the previous base model, we just use the same methodology to do our mathematical analysis. We use the theory of equilibrium point as well, which has the equation:

$$\begin{aligned} S_{t+1} &= S_t = S^* \\ F_{t+1} &= F_t = F^* \\ I_{t+1} &= I_t = I^* \\ R_{t+1} &= R_t = R^* \end{aligned}$$

And thus we can compute and find the equilibrium point of this system of model. It should be

$$S^* = 1, \quad I^* = 0, \quad F^* = 0, \quad R^* = R^*$$

We have the same conclusion to the base model, the model suggests that, over time, the system will asymptotically approach a disease-free equilibrium. Then look back on the previous one, it is time to do the Jacobian related analysis.

$$\begin{aligned} f_1 &= S - \beta SI - \beta_F SF + \mu - \mu S \\ f_2 &= I + \beta SI + \beta_F SF - \gamma I - \mu I - \epsilon I \\ f_3 &= F + \gamma I - \sigma F \\ f_4 &= R + \sigma F + \epsilon I \end{aligned}$$

The Jacobian Matrix is given by:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial S} & \frac{\partial f_1}{\partial I} & \frac{\partial f_1}{\partial F} & \frac{\partial f_1}{\partial R} \\ \frac{\partial f_2}{\partial S} & \frac{\partial f_2}{\partial I} & \frac{\partial f_2}{\partial F} & \frac{\partial f_2}{\partial R} \\ \frac{\partial f_3}{\partial S} & \frac{\partial f_3}{\partial I} & \frac{\partial f_3}{\partial F} & \frac{\partial f_3}{\partial R} \\ \frac{\partial f_4}{\partial S} & \frac{\partial f_4}{\partial I} & \frac{\partial f_4}{\partial F} & \frac{\partial f_4}{\partial R} \end{bmatrix}$$

Thus, we use the similar method to deal with this Jacobian matrix and get similar eigenvalues to the base model. These four eigenvalues are solved by this equation.

$$\begin{aligned} \det(J - \lambda I) &= (1 - \mu - \lambda)(1 - \lambda) \\ &\quad \times [(1 + \beta - \mu - \gamma - \epsilon - \lambda) \\ &\quad \times (1 - \sigma - \lambda) - \beta_F \cdot \gamma] \end{aligned}$$

Building on the basic model, the extension of the SIFR model introduces an intervention parameter(ϵ), which adds complexity to the system and makes the direct determination of the dominant eigenvalue more challenging. Due to the differing effects of the parameters on the system, the exact value of the dominant eigenvalue must be determined based on specific parameter settings, including(ϵ).

By setting initial parameter ranges and analyzing the model, some preliminary insights can be obtained. According to stability theory, if the dominant eigenvalue is

equal to 1, the trajectory of the system becomes uncertain. Therefore, the extended SIFR model requires computational simulations to further understand its dynamic behavior. On the other hand, if the dominant eigenvalue is greater than 1, it indicates that the equilibrium point of the extended model is unstable. This implies that, without effective interventions and adjustments, the system could become unstable, potentially leading to uncontrolled spread of the outbreak.

3.4 Extended R Simulation Analysis

In our extended model, we construct an environment which is closer to the real world scenario. We still research the parameters sigma and beta to test our previous conclusion and get inspiration on how to constrain Ebola transmission with effective policies.

3.4.1 First R Simulation Analysis

As the base model, we firstly focus on how sigma (rate of burials within a certain period) affects change of various populations. If we increase sigma from 0.02 to 0.04, the peak percentage of infected individuals decreases slightly, although this change is difficult to observe with the naked eye. In the long run (after day 180), we can see a larger percentage of susceptible population (healthy population), meaning replacing traditional, long-lasting burial with efficient on-site burial can effectively limit the spread of pandemic. The result also indicates that the percentage of fatalities in the population encounters a significant drop. Ideally, it is important that citizens believe in government and follow immediate burial; the action of effective burial results in positive consequences in both short run (before day 180) and long run (after day 180). In the short run, we can see less proportion of the population getting infected without being hospitalized and getting infected due

to joining in traditional burial; while in the long run, we can see more survivors after pandemic.

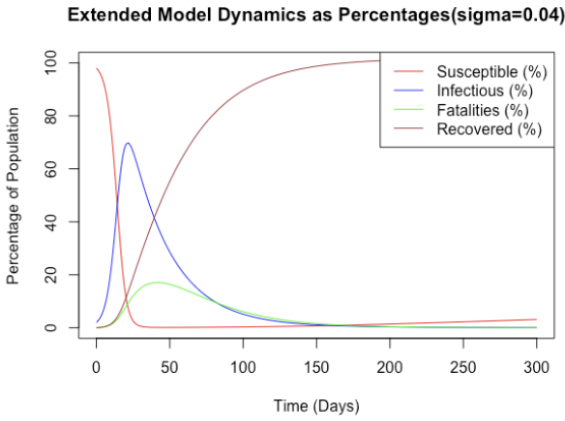
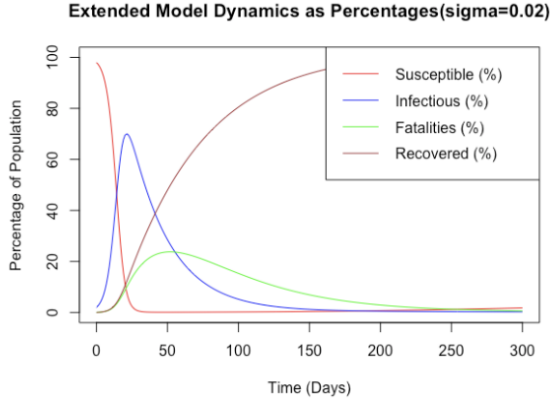


Figure 5: Extended Model Simulation Result based on Changes in Burial Efficiency

As the base model, we firstly focus on how sigma (rate of burials within a certain period) affects change of various populations. If we increase sigma from 0.02 to 0.04, the peak percentage of infected individuals decreases slightly, although this change is difficult to observe with the naked eye. In the long run (after day 180), we can see a larger percentage of susceptible population (healthy population), meaning replacing traditional, long-lasting burial with efficient on-site burial can effectively limit the spread of pandemic. The result also indicates that the percentage of fatalities in the population encounters a significant drop. Ideally, it is important that citizens believe in government and follow immediate

burial; the action of effective burial results in positive consequences in both short run (before day 180) and long run (after day 180). In the short run, we can see less proportion of the population getting infected without being hospitalized and getting infected due to joining in traditional burial; while in the long run, we can see more survivors after pandemic.

3.4.2 Second R Simulation Analysis

For the parameter beta (infection transmission rate), we see that the major difference between the two scenarios, which beta values equal 0.000318 and 0.000518 respectively, is that the proportion of infectious population gets larger if we raise the beta. Similar to performance in base model, the proportion distribution for all kinds of population assimilates in the long run (after day 180). But the R simulation for the extended model again shows that the infection transmission rate has a significant impact on the short term situation. If the pandemic becomes more aggressive and affects the population quicker, we can see from the graph that the percentage of infectious population increases faster and gets to its peak faster. (at about day 25 when beta = 0.000318 and at about day 20 when beta = 0.000518). Also, if beta increases by 0.0002, the percentage of susceptible population, or the healthy population who have not got infected, also reduces sooner. In the real world, the rapidly growing number of patients will definitely pose a challenge to the capacity of the healthcare system. Therefore, based on R simulation, we still recommend the local governments taking action to isolate individuals to prevent the rapid spread of the virus.

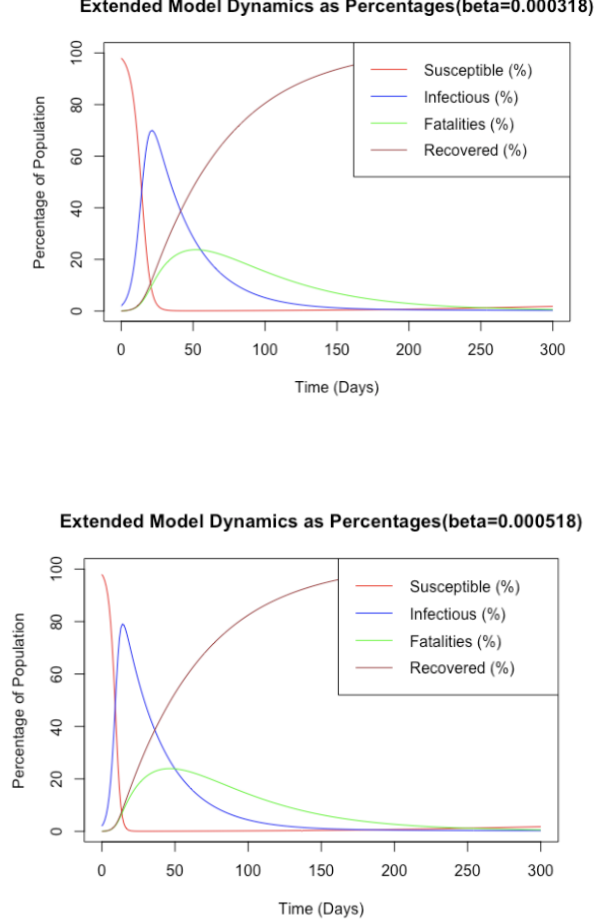


Figure 6: Extended Model Simulation Result based on Changes in Infection Transmission Rate

4 Discussion and Analysis of Research Questions

After the detailed mathematical analysis and the verification from R simulation. We have had the clear outline for this type of virus, Ebola. We would like to answer some research questions we have mentioned at the beginning of the paper. Those questions assist us in finishing some suggestions or in understanding the virus itself.

Research Question 1: What parameter values can we use to monitor the infection?

The key point for this modeling procedure is the parameters. SIFR model relied on

them to do the adjustment, including monitoring. Among them, the infection rate β plays a central role in determining the transmission of the disease through susceptibility and the interaction between infected individuals. Similarly, the funeral transmission rate β_F is particularly important in the social-cultural context, because traditional burial customs involve close contact with the deceased. The burial rate is another key parameter that reflects the efficiency of funeral-related interventions. In addition, γ captures the transition rate from infection to death, while μ takes into account the birth rate and natural mortality rate, assuming that the population is stable. The extended model introduces ϵ , indicating immediate burial without funerals, which provides a realistic approximation for public health strategies aimed at minimizing secondary transmission. The simulation results emphasize the importance of these parameters. For instance, the increase of σ leads to a significant decrease in the infection rate, we have shown that in Figure 4. In addition, the higher ϵ reduces the transmission potential of the dead waiting for burial, thus improving the overall effectiveness of the intervention strategy. These parameters provide a detailed framework for monitoring the progress of the epidemic and formulating targeted public health response measures.

Research Question 2: How do funerals contribute to the spread of the Ebola virus within populations?

How do funerals lead to the spread of the Ebola virus among crowds? Funerals are the main approach of transmission of Ebola, particularly in areas where traditional ceremonies involve mass processing of the dead. In our underlying model, β_F directly quantifies such risk. On the one hand, the R simulation results showed that the infection curve became steeper with increasing β_F , which means a faster and wider epidemic development. On the other hand, higher values of σ i.e. reflect better burial man-

agement can effectively mitigate these risks. For instance, a comparison with $\sigma = 0.02$ and $\sigma = 0.04$, which means a significant reduction in the number of secondary infections caused by embalming. The extended model further adds depth by incorporating ϵ i.e. which stands for immediate and safe burial practices. This adjustment not only decreases the number of infections that are related to funerals but also increases the proportion of the healthy population in the long run. The R simulation results give us that incorporating into burial protocols can lead to a more sustainable recovery phase, minimizing the impact of epidemics. In addition to the numerical results, this highlights the need for culturally sensitive interventions to balance public health priorities and community traditions.

Research Question 3: What is the impact of public health interventions in reducing transmission in funerals on epidemic control?

Public policy intervention is a critical and necessary approach to controlling the spread of specific funerals and the overall spread of the epidemic. Our paper results show that promoting immediate burials through the development of reasonable burial procedures is an effective strategy to reduce secondary infections, which we have shown in the previous R Simulation results. For instance, increasing σ can reduce the peak of infection and shorten its reflecting time, as shown in Figure 4. The results of the mathematical analysis also highlight the importance of reducing the overall infection rate β through public health and non-treatment perspective such as isolation, contact tracing, and public education. When β is kept at a low level, the epidemic

spreads more slowly, which is good. On the contrary, an increase in β will cause the infection curve to rise more rapidly, which has a significant impact on the efficiency of public health infrastructure. We generally believe that a more relaxed environment can produce higher efficiency. Community participation and education are key to the implementation of our assumptions above. By integrating this information, it will be very effective in controlling the spread in the long term.

5 Conclusion

We have completed a very lengthy mathematical analysis and obtained very satisfactory results. Unfortunately, the situation in the real world is often more complicated than we imagined. All the assumptions we set may not hold true. For this reason, we may need to introduce many more variables to meet the conditions of the model. Our results are not necessarily accurate, but we have taken into account the most likely situations in the real world. Although we have not conducted an accuracy analysis on the statistical model and simulation, we can still say that the above results can be put into the real society, especially for the analysis of the expanded model, which will be more robust.

In addition, this statistical method can also be used to analyze other infectious diseases, including the new crown epidemic, but more parameters and assumptions may be added. Finally, we hope to apply the policy recommendations obtained from this analysis in real society to increase the welfare of human society.

6 R codes

Base model R simulation

```
1 # Total Population
2 N <- 1000 # Total population size
3
4 # Initial Conditions
5 IO <- 20 # Initial number of infectious individuals
6 FO <- 0 # Initial number of fatalities
7 RO <- 0 # Initial number of recovered individuals
8 SO <- N - IO - FO - RO # Susceptible population is the remainder
9
10 # Model Parameters
11 beta <- 0.000318 # Infection transmission rate
12 beta_F <- 0.0005 # Infection transmission from fatalities
13 sigma <- 0.02 # Death rate from infections
14 mu <- 0.0002 # Natural birth/death rate
15 gamma <- 0.0175 # Recovery rate
16 epsilon <- 0.0175 # Rate of deaths without funerals
17
18 # Time Steps
19 times <- seq(0, 300) # Define the time sequence from 0 to 300 (
# discrete time steps)
20
21 # Initialize State Matrix
22 # Rows represent: Susceptible (S), Infectious (I), Fatalities (F),
# Recovered (R)
23 # Columns represent each time step
24 state_matrix <- matrix(NA, nrow = 4, ncol = length(times))
25 state_matrix[, 1] <- c(SO, IO, FO, RO) # Set initial values for the
# first time step
26
27 # Loop Through Time Steps to Update the SIRD Model
28 for (t in 2:length(times)) {
29 # Retrieve Values from the Previous Time Step
30 S <- state_matrix[1, t - 1] # Susceptible population at time t-1
31 I <- state_matrix[2, t - 1] # Infectious population at time t-1
32 F <- state_matrix[3, t - 1] # Fatalities at time t-1
33 R <- state_matrix[4, t - 1] # Recovered population at time t-1
34
35 # Differential Equations for SIRD Dynamics
36 dS <- -beta * S * I - beta_F * S * F + mu * N - mu * S # Change in
# Susceptible
37 dI <- beta * S * I + beta_F * S * F - gamma * I - epsilon * I - mu *
# I # Change in Infectious
38 dF <- gamma * I - sigma * F # Change in Fatalities
39 dR <- sigma * F + epsilon * I # Change in Recovered
40
41 # Update the States for Current Time Step
42 state_matrix[1, t] <- S + dS # Update Susceptible
43 state_matrix[2, t] <- I + dI # Update Infectious
44 state_matrix[3, t] <- F + dF # Update Fatalities
45 state_matrix[4, t] <- R + dR # Update Recovered
46 }
47
48 # Convert States to Percentages of Total Population
```

```

49 percent_matrix <- (state_matrix / N) * 100 # Convert absolute numbers
    to percentages
50
51 # Plot Results
52 plot(
53   times, percent_matrix[1, ], type = "l", col = "red",
54   xlab = "Time(Days)", ylab = "Percentage of Population",
55   main = "SIRF Model Dynamics as Percentages", ylim = c(0, 100)
56 )
57 lines(times, percent_matrix[2, ], col = "blue") # Infectious
58 lines(times, percent_matrix[3, ], col = "green") # Fatalities
59 lines(times, percent_matrix[4, ], col = "brown") # Recovered
60
61 # Add a Legend to Describe Each Curve
62 legend(
63   "topright",
64   legend = c("Susceptible(%)", "Infectious(%)", "Fatalities(%)", "
        Recovered(%)"),
65   col = c("red", "blue", "green", "brown"), lty = 1
66 )

```

Extended model

```

1 library(dplyr)
2 library(tidyr)
3 library(ggplot2)
4
5 # Parameters
6 N <- 1000 # Total population
7 S0 <- 900 # Initial susceptible population
8 IO <- 100 # Initial infectious population
9 FO <- 0   # Initial fatalities
10 RO <- 0   # Initial recovered
11 beta <- 0.000318 # Infection transmission rate
12 beta_F <- 0.0005 # Infection transmission from fatalities
13 gamma <- 0.0175 # Recovery rate
14 sigma <- 0.02 # Death rate from infection
15 mu <- 0.0006 # Birth/death rate
16
17 # Time steps
18 times <- seq(0, 400, by = 1)
19
20 # Initialize data frame to store S, I, F, R
21 sirf_data <- tibble(
22   time = times,
23   S = NA_real_,
24   I = NA_real_,
25   F = NA_real_,
26   R = NA_real_
27 )
28
29 # Initial values
30 sirf_data <- sirf_data %>%
31   mutate(
32     S = if_else(time == 0, S0, NA_real_),
33     I = if_else(time == 0, IO, NA_real_),
34     F = if_else(time == 0, FO, NA_real_),
35     R = if_else(time == 0, RO, NA_real_)
36 )

```

```

37
38 # SIRF Model Dynamics
39 for (t in 2:length(times)) {
40   prev <- sirf_data[t - 1, ]
41
42   dS <- -beta * prev$S * prev$I - beta_F * prev$S * prev$F + mu * N -
      mu * prev$S
43   dI <- beta * prev$S * prev$I + beta_F * prev$S * prev$F - gamma *
      prev$I - sigma * prev$I - mu * prev$I
44   dF <- gamma * prev$I - sigma * prev$F
45   dR <- sigma * prev$F
46
47   sirf_data <- sirf_data %>%
48     mutate(
49       S = if_else(time == t - 1, prev$S + dS, S),
50       I = if_else(time == t - 1, prev$I + dI, I),
51       F = if_else(time == t - 1, prev$F + dF, F),
52       R = if_else(time == t - 1, prev$R + dR, R)
53     )
54 }
55
56 # Convert to long format for plotting
57 sirf_long <- sirf_data %>%
58   pivot_longer(cols = c(S, I, F, R), names_to = "category", values_to =
      "population") %>%
59   mutate(percentage = (population / N) * 100)
60
61 # Plot results
62 ggplot(sirf_long, aes(x = time, y = percentage, color = category)) +
63   geom_line() +
64   labs(
65     title = "SIRF Model Dynamics (beta = 0.000318)",
66     x = "Time",
67     y = "Percentage of Population"
68   ) +
69   scale_color_manual(values = c("red", "blue", "green", "brown")) +
70   theme_minimal()

```

References

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